

REPRO Report

[Evidence guided] Persistent Datapatch bug

Case md-12620-terra-overnight-02 | Mindustry | Awaiting human
Revision: 63b2069e9685063c74abc41fae84a66f73e9e09b
Reported platform: Android (reported); Linux (planned evaluation) | Build: Unspecified
Generated 2026-09-14 19:16 UTC from the current saved case.

Candidate ready for review. Validation gates satisfied; inspect any pre-existing test failures.

Reproduction	Time to first proof	Replay actions
5/5 clean runs	9:31	31 to 30

Player report

Platforms

Android

Build

BE 27790

Issue

When you save a datapatch in a map, suddenly, attempting to remove it does not work.
When you remove it, it gets temporarily removed, but gets added back when you open the map.
I tested this with or without mods and it still happened by the way.

Steps to reproduce

Make map
Add data-patch
Save map
Delete data-patch
Save map
Leave editor
Come back to map editor
See that the data-patch is back.

Mods used

I tested it in vanilla but here are the mods I usually use.

Auto Research (gone from in-game mod browser?)
Patch Editor (I think this mod doesn't cause the issue)
Patch Viewer
CLaJ

Save file

I zipped the msav here

[persistent datapatch bug.zip](https://github.com/user-attachments/files/32028828/persistent.datapatch.bug.zip)

(Crash) logs

No response

REPRO recording guidance (not part of the upstream report)

A previous local attempt reached the reported sequence, but the final oracle omitted its recorded initial-save screenshot. Two of five independent replay judgments were inconclusive because that prerequisite was not visible in the selected evidence. This is a fresh attempt; those prior judgments do not establish the current result.

Capture and select the important checkpoints in chronological order: the map identity and existing patch, visible successful initial save, confirmed deletion with an empty list, visible successful save after deletion, exit/reopen of the same map, and the patch list after reopening. Include both successful-save screenshots in the final sequence oracle,

within its eight-checkpoint limit. Use neutral labels for the resulting state; inspect what is actually visible rather than assuming disappearance or reappearance. If prerequisites cannot be demonstrated, report that limitation.

Expected behavior: After a datapatch is deleted and the map is saved, reopening the same map should retain the deletion.

Initial questions

Recorded at intake; later findings and validation may address these questions.

- The reported behavior has not been independently verified.
- It is unknown whether the linked ZIP contains the relevant map save, a reproducible starting state, or valid/unmodified data.
- It is unknown whether the map was saved successfully both before deletion and after deletion.
- It is unknown whether the same map was reopened after leaving the editor.
- It is unknown whether the datapatch list was empty immediately after deletion or whether the apparent removal affected only the current UI state.
- The exact Mindustry version/commit corresponding to build BE 27790 is not established; a target commit was supplied separately but not confirmed by the report.
- The report names Android, while Linux is listed as a planned evaluation platform; cross-platform applicability is unknown.
- The vanilla test configuration, mod disablement procedure, and whether a restart occurred between tests are not specified.
- The listed mods' versions and whether any were enabled during the reported failing run are unknown.
- No crash log was provided; it is unknown whether any errors or warnings occurred during saving or reopening.

Recorded reproduction

1. Open map editor from main menu
2. Create a new editable map
3. Focus map-name text field
4. Enter unique map name
5. Create named test map
6. Open editor menu
7. Open map information
8. Open Data Patches and Assets
9. Add a datapatch
10. Load datapatch from current clipboard
11. Return to map information with datapatch added
12. Open menu to save map containing datapatch
13. Save map with added datapatch
14. Open editor menu to remove datapatch
15. Open map information
16. Inspect persisted datapatch before deletion

17. Request deletion of saved datapatch
18. Confirm datapatch deletion
19. Leave empty datapatch list after deletion
20. Open editor menu after deletion
21. Save map after datapatch deletion
22. Open editor menu to leave saved map
23. Quit editor and return to map list
24. Confirm leaving editor after saving deletion
25. Open map editor map list
26. Reopen the same custom map after saved deletion
27. Open named map in editor for post-reopen inspection
28. Open reopened map editor menu
29. Open reopened map information
30. Inspect datapatches after reopening same saved map

Source diagnosis

The confirmed deletion path removes the PatchAsset only from DataManager's per-type patch Seq and rebuilds the UI. It does not call `DataManager.reloadPatches` (or otherwise rebuild `orderedAssets`). Map saving serializes `orderedAssets` via `getAllAssets()`, so it can write the stale pre-deletion asset list. On reopen, `readDataPatches` loads that serialized patch again.

`core/src/mindustry/editor/data/MapPatchesView.java`

This is the most direct source of the reported UI-versus-persisted-state split: the visual list is rebuilt from the now-empty per-type collection, but `DataManager`'s save-facing aggregate can remain stale.

`core/src/mindustry/mod/DataManager.java`

This establishes the stale-cache mechanism connecting the missing reload in the deletion handler to persistence. It also explains why adding/replacing, which calls `reloadPatches`, would not have the same aggregate-list staleness.

`core/src/mindustry/io/SaveVersion.java`

This is the persistence endpoint that makes the in-memory aggregate discrepancy observable after save, exit, and reopen. It is not itself shown to mutate incorrectly; it faithfully serializes its supplied aggregate.

`core/src/mindustry/editor/data/MapBundlesView.java`

This is corroborating evidence of a broader editor asset-removal pattern and may imply an analogous persistence risk for bundles. It is lower ranked because the reported replay deletes a datapatch, not a bundle.

Proposed patch

Why this patch should work

Cause: the delete confirmation at MapPatchesView.java:84-89 removes the PatchAsset from the mutable per-type sequence returned by `state.data.getPatches()` but only rebuilds the dialog. `DataManager` maintains a separate `orderedAssets` aggregate: `reloadPatches()` rebuilds it (DataManager.java:169-176), while `getAllAssets()` returns that aggregate directly (DataManager.java:263-265). Saving serializes `getAllAssets()` (SaveVersion.java:595-601), so without a reload the save can still contain the deleted patch and loading restores it.

The added line calls the same patch reload path already used after add/replace in MapPatchesView.java:125-143. It unapplies the old patch effects, applies the remaining patches, and rebuilds `orderedAssets`. Thus the save-facing aggregate no longer includes the deleted PatchAsset before the UI is rebuilt, so a subsequent map save should persist the deletion.

This is a one-line, datapatch-only change in the confirmed deletion handler; it does not alter serialization, save format, the regression oracle, or unrelated asset behavior.

Risks and tradeoffs

- Deleting a patch now immediately reapplies all remaining patches. This is consistent with add/replace behavior, but reviewers should verify that removing a patch is intended to revert its runtime content effects immediately, not only update persisted editor state.
- `reloadPatches` may have a perceptible cost or expose errors from remaining malformed patches at deletion time; this follows the established add/replace path and is preferable to retaining stale serialized state.
- Other asset editors may have analogous stale-aggregate deletion paths, but they are deliberately out of scope for this datapatch replay and are not changed.

Code changes

Lines starting with - are removed; lines starting with + are added. Unmarked lines provide context.

```
diff --git a/core/src/ministry/editor/data/MapPatchesView.java b/core/src/ministry/editor/data/MapPatchesView.java
index b814576..b53bef9 100644
--- a/core/src/ministry/editor/data/MapPatchesView.java
+++ b/core/src/ministry/editor/data/MapPatchesView.java
@@ -84,6 +84,7 @@ public class MapPatchesView implements AssetView{
    list.button(Icon.trash, Styles.graySquarei, iconMed, () -> {
        ui.showConfirm("@patch.delete.confirm", () -> {
            patches.remove(patch);
+           state.data.reloadPatches(patches);
            diag.rebuild();
        });
    }).size(h);
```

Validation results

All five required checks passed. Approval is a separate human review decision.

Check	Result	Recorded evidence
Regression before patch	PASS	Bug oracle triggered in 5/5 clean runs; the regression therefore fails on the pre-fix build.
Candidate build	PASS	Build exit code 0
Existing tests	PASS	Test exit code 0.
Original replay after patch	PASS	5/5 reached expected state without the symptom; bug seen 0 times.
Smoke test	PASS	Clean desktop launch and live process after startup; deeper gameplay smoke coverage is not implemented.

Source analysis limitations

These are the source reviewer's recorded notes. That stage inspects code; see Recorded reproduction and Validation results for executed game checks.

- Only source was inspected; no game client was run and no supplied screenshot, ZIP, map-save bytes, or runtime logs were available through the tools.
- The report identifies Android/build BE 27790 while source provenance relative to the replay commit was not independently established. The control/data flow in the inspected source nevertheless directly matches the supplied deterministic replay.
- This localizes a cause consistent with the report, but does not determine the future human fix or prove whether other asset types are intended to refresh their runtime loaders on deletion.

Usage and review

Each replay starts with a fresh profile. Reproduction counts describe this case; action reduction is bounded and is not proof of a globally minimal replay.

80 model calls; 926,267 input tokens; 15,071 output tokens. Usage accumulates across jobs on this case. PDF export uses saved data and makes no model calls.

The proposed patch lives in a disposable game workspace. Handoff approval records a local review decision and does not publish or merge changes upstream. Visual checks are model-based; startup smoke does not establish broad gameplay coverage.

Investigation timeline

All timestamps are UTC. First and latest entries are saved event times, not active work durations. Repeated attempts and pauses are retained. Selected milestones appear below; the workspace has the full searchable log.

Triage / 4 records

First entry: 14 Sep 2026 16:45:33 UTC
Latest entry: 14 Sep 2026 17:00:32 UTC

14 Sep 2026 16:45:33 | Fresh evidence-guided attempt; original case and its 3/5 result retained separately. Guidance addresses checkpoint selection only.

14 Sep 2026 17:00:32 | Reported persistent datapatch deletion issue: after deleting a datapatch from a saved map and reopening the map editor, the datapatch is reported to be present again.

Reproduce / 259 records

First entry: 14 Sep 2026 17:00:32 UTC
Latest entry: 14 Sep 2026 17:17:49 UTC

14 Sep 2026 17:00:32 | Fetching the exact revision and preparing build dependencies.

14 Sep 2026 17:17:49 | Symptom observed. The saved map's Patches UI shows one patch before deletion, then explicitly shows no patchsets after deletion. After a visible save, exit/reopen of the map named DatapatchPersistenceTest, the Patches UI again...

Reduce / 407 records

First entry: 14 Sep 2026 17:17:49 UTC
Latest entry: 14 Sep 2026 17:37:07 UTC

14 Sep 2026 17:17:49 | Proposing action deletions and accepting them only after fresh baseline replays.

14 Sep 2026 17:37:07 | Verified in 5/5 replays; 31 30 actions.

Localize / 28 records

First entry: 14 Sep 2026 17:37:07 UTC
Latest entry: 14 Sep 2026 17:37:55 UTC

14 Sep 2026 17:37:07 | Tracing observed behavior into the pre-fix source.

14 Sep 2026 17:37:55 | Saving the confirmed replay as an executable regression test.

Validate / 177 records

First entry: 14 Sep 2026 17:37:55 UTC
Latest entry: 14 Sep 2026 17:49:05 UTC

14 Sep 2026 17:37:55 | Designing a small candidate diff in the disposable repository.

14 Sep 2026 17:49:05 | Expected state reached; symptom absent.

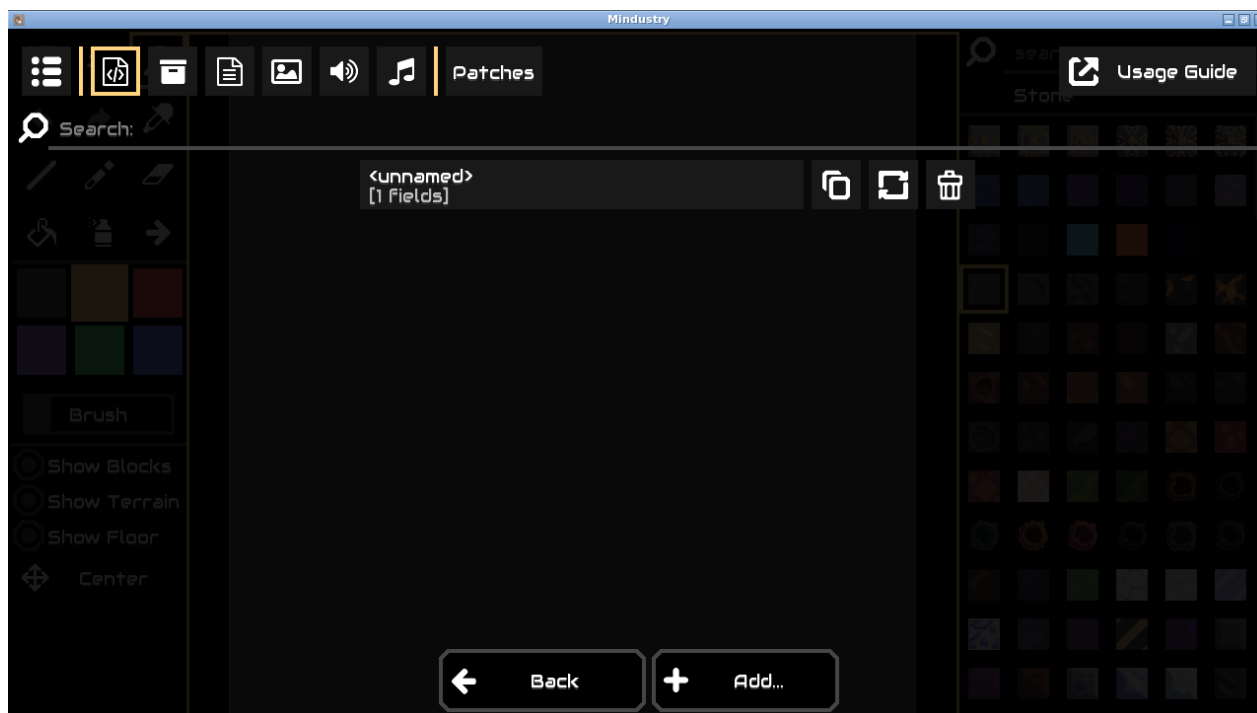
Review / 1 records

First entry: 14 Sep 2026 17:49:16 UTC
Latest entry: 14 Sep 2026 17:49:16 UTC

14 Sep 2026 17:49:16 | Candidate ready for review. Validation gates satisfied; inspect any pre-existing test failures.

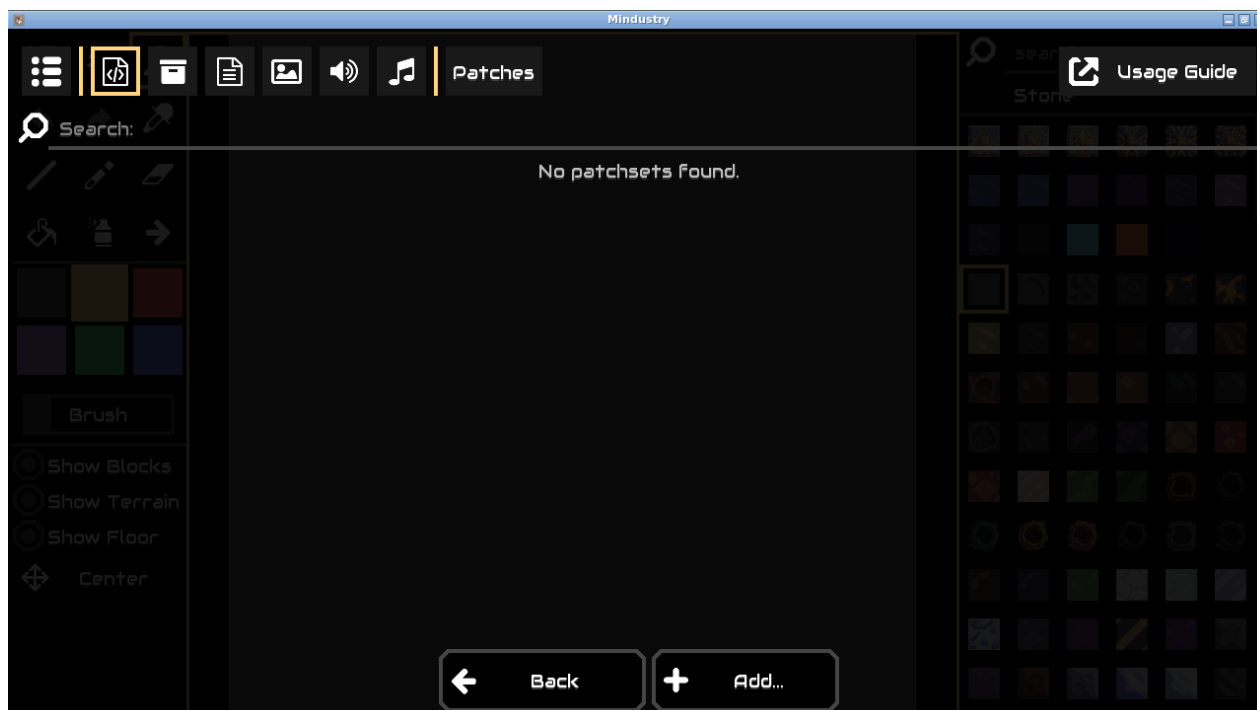
Recorded screenshots

Baseline replay evidence



md-12620-terra-overnight-02-6020ead1fd0b0e515c4f-reduced-confirmation.png

Candidate replay outcome



md-12620-terra-overnight-02-2e0253837f7c1f5328b7-post-patch.png